Don't-care

Real specs have don't-cares , CEC must only compare care inputs, not illegal or unconstrained bits

The idea

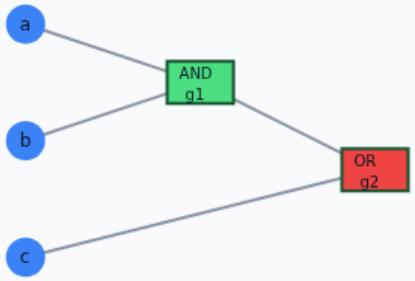
- Dont_care_mask with c fixed to 0 still finds a=1,b=1,c=0 mismatch on impl_bad

Care mask

DON'T-CARE

Care mask

Step 1 / 5 · mask · module03-01-dont-care



Explanation

Don't-cares mark PIs as X (−1) or fixed 0/1.

- X bits

```
care set
```

Care mask

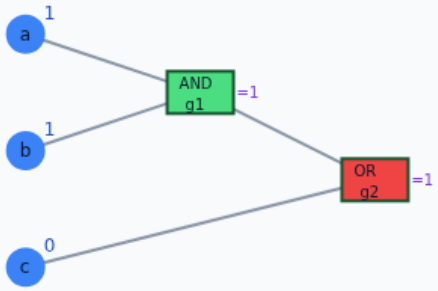
Fix c=0

DON'T-CARE

Fix c=0

Step 2 / 5 · fix-c · module03-01-dont-care

{"equivalent":false,"rowsChecked":4,"counterexample":{"a":1,



Explanation

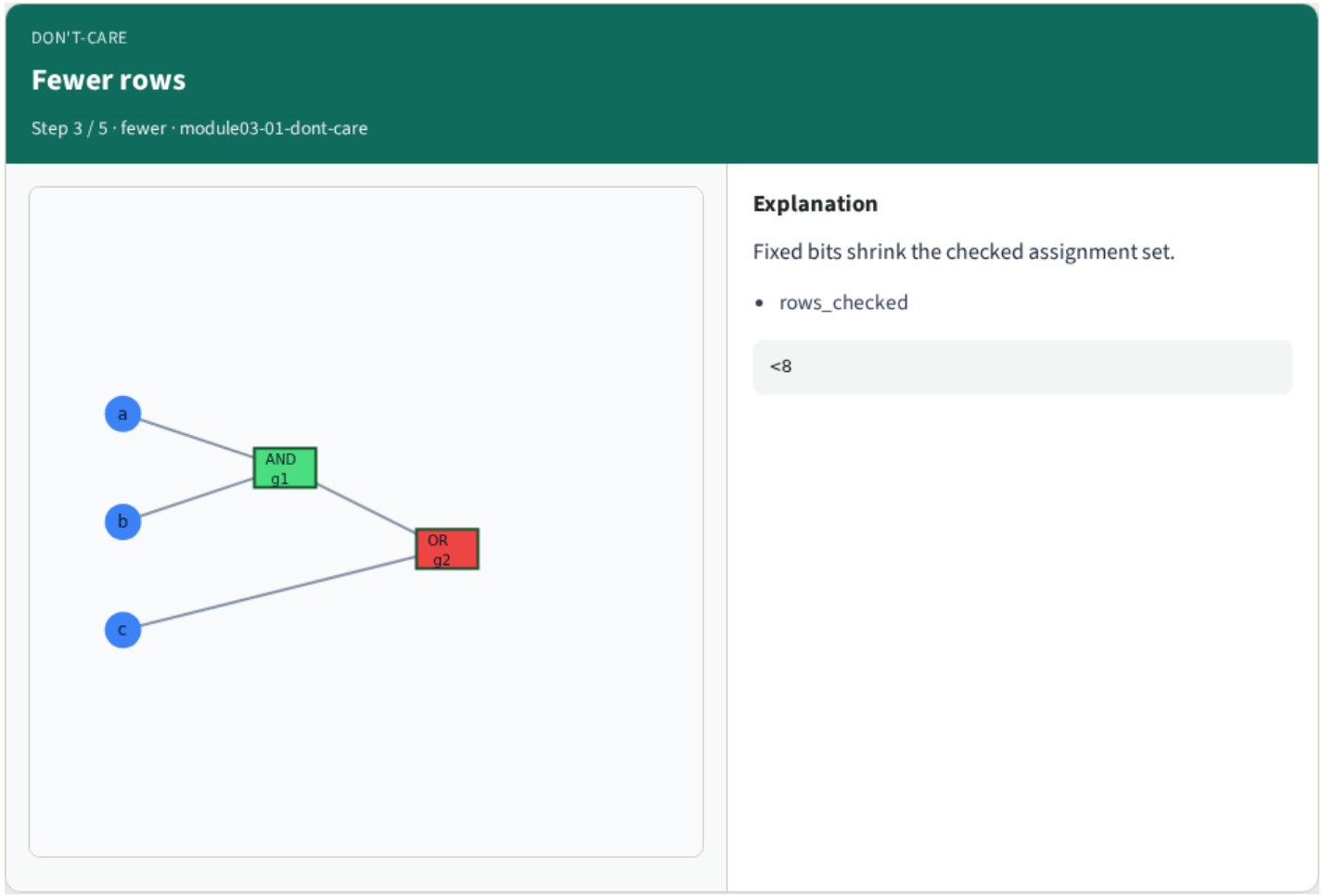
Mask with c fixed to 0 still catches the bad cex.

- partial

```
still fail
```

Fix c=0

Fewer rows



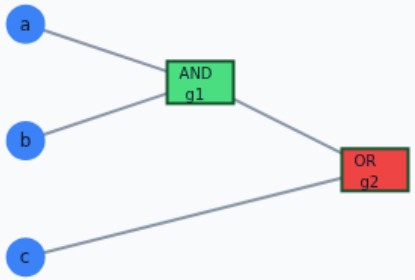
Fewer rows

X means ignore

DON'T-CARE

X means ignore

Step 4 / 5 · x-ok · module03-01-dont-care



Explanation

X on a bit means both 0/1 are allowed for that PI.

- don't care

```
spec care
```

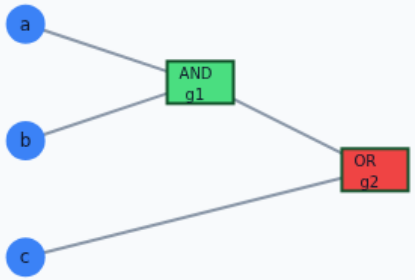
X means ignore

Real DC

DON'T-CARE

Real DC

Step 5 / 5 · use · module03-01-dont-care



Explanation

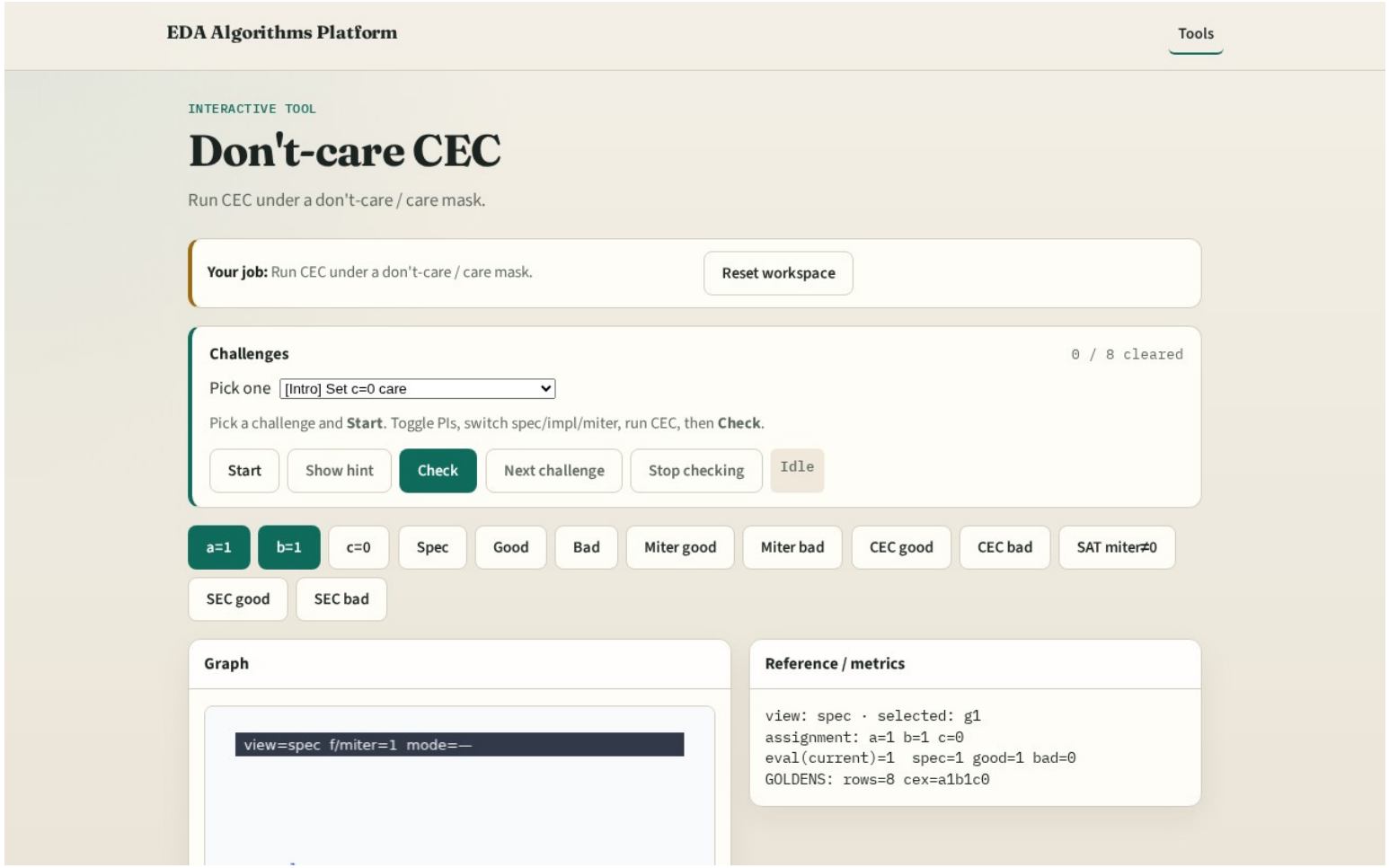
Synthesis don't-cares feed CEC care sets in industrial flows.

- ODC/SDC

```
Next SEC
```

Real DC

Browser lab track



Browser lab starter

Implement track

- Call dont_care_mask(spec, impl_bad, {'a':-1,'b':-1,'c':0})

Pitfalls

- Treating X as 0
- Checking illegal input combinations as failures

Your turn

- Next: sequential unrolling